

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

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4 Implementation and Empirical Results

Chapter 3 specified how EndorseRank and the Adaptive Weighted PageRank (AWP) baseline are built, which construct checks are applied, how uncertainty is quantified and how runtime is timed. This chapter applies that design to Arbitrum One and reports what was observed. It is organized as a single case study: one chain, one observation window, one matched cohort for all same-window comparisons, one spender cohort for the temporal holdout and one expanded address pool for node-count sensitivity. Tables are generated from the evaluation summary in Section 3.13, and confidence intervals accompany every Kendall τ coefficient.

4.1 Case study setting: Arbitrum One

Arbitrum One was selected as the case because it is a high-activity Ethereum rollup with mature DeFi protocols (Section 2.2), its logs are available in the public BigQuery export used by the pipeline, and a margin-trading protocol on it (GMX V2) supplies the wallet seed set described below. The observation window runs from 1 December 2025 to 31 May 2026. This window is six months, which is long enough to encompass multiple market regimes and allow the tenure and active-month proxies to vary, while also short enough for the latest-allowance snapshot to remain current relative to the transfer history AWP uses.

4.1.1 Cohorts

There are three sets of addresses, each serving a specific purpose.

- Matched cohort ($n = 5,521$). The GMX V2 margin-trading wallets are the ones that are connected, with non-zero weights, in both G_{approve} , the latest-allowance endorsement graph, and G_{transfer} , the time-decayed transfer graph over the window. All results in the same-window

alignment, difference-test and benchmark sections refer to this cohort. By requiring connectivity in both graphs we ensure that neither method is evaluated on wallets that can only produce the teleportation floor score from it.

- Holdout spender cohort ($n = 1,335$). Wallets that appear as spenders in at least one allowance over the scoring period (all events up to 28 February 2026). These scores are evaluated against owner–spender approval pairs and transfer senders that were first observed over the holdout period (1 March 2026 to 31 May 2026) (Section 3.7).
- Expanded wallet pool ($N = 99,080$). The matched cohort is combined with 93,559 additional addresses that are sampled from the monthly ERC-20 activity on the chain. This pool is only used for node-count sensitivity of runtime and sample-definition sensitivity (Section 3.12.1); the caveat that these supplemental addresses contribute nodes but no additional edges is mentioned in the relevant results section.

The sizes of the graphs for the matched cohort are provided in Table 4.1 alongside the benchmark. The endorsement graph has approximately one-ninth the number of edges of the transfer graph on the same wallets. This imbalance is due to the nature of the two graphs rather than any error in their construction, as they both originate from the same set of wallets and window. EndorseRank only retains the latest decimal-adjusted allowance for each owner–spender pair, so multiple approvals result in a single directed edge, while AWP sums all transfers between two wallets into a time-decayed weight. Any differences in the scores and alignment between the two methods are due to differences in ranking logic and graph semantics rather than inconsistent sampling.

4.1.2 Implementation

All the algorithms use the same implementation of PageRank in Python, with $d = 0.85$, tolerance $= 10^{-8}$ and maximum iterations; the single-layer methods only differ by their edge list, and the coupled operator simply combines the two edge lists per node according to Equation (3.6) (Section 3.15). Scores, statistics by proxy, confidence intervals from bootstrapping and timing are saved to one file per algorithm run; the table shown below is from the run dated 14 September 2026. The

repository revision, config file and summary files of that run are provided in Appendix 8.1.

4.2 Computational benchmark (Claim C1)

Claim C1 states that EndorseRank is computationally less expensive than AWP on the same input. Table 4.1 lists the PageRank solve time, its standard deviation over the five timed repeats, the peak memory usage of the warm-up solve, the number of iterations until convergence, and the size of the evaluation subgraph of each method, on the matched cohort, as per the timing procedure of Section 3.11.

Table 4.1: Computational benchmark on the matched wallet cohort ($n = 5,521$; AWP/EndorseRank runtime ratio $10.2\times$). Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs. $|V|$, $|E|$ = nodes and edges of each method’s evaluation subgraph. C-PR walks on the union of both layers, so $|E|$ is the sum of the two edge sets; S-PR runs on the transfer layer and its runtime includes the EndorseRank solve that supplies its restart vector.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$ V $	$ E $
EndorseRank	0.401	0.016	4.6	37.0	6,442	14,727
AWP	4.105	0.193	38.2	100.0	30,791	137,087
C-PR ($\lambda = 0.5$)	4.760	0.471	41.3	76.0	31,242	151,814
S-PR	4.156	0.757	39.1	145.0	30,791	137,087

There are three reasons why the runtime ratio in the table is as it is. First, both stages of the timed solve, constructing the sparse transition matrix from the edge list, and performing each power-iteration step, are linear in $|E|$ (see Appendix 8.3.3), and the endorsement graph contains approximately one-ninth as many edges as the transfer graph. Second, EndorseRank converges in approximately one-third the number of iterations as AWP; the transfer operator, with its long-tailed decaying weights, is slower to mix. Third, peak memory also scales with $|E|$, since adjacency, weight and residual buffers are all proportional to the size of the edge list. The standard deviations indicate that the noise from run to run is about an order of magnitude lower than the difference between the methods.

The two hybrid rows illustrate what the speed result covers and what it does not. C-PR walks on the union of the two edge sets (151,814 edges, 31,242 nodes), and its solve time is 4.76 s, which is marginally greater than the AWP time; it converges in fewer iterations than AWP (76 vs. 100), but each iteration touches more edges. S-PR runs on the transfer layer, and takes 4.16 s including the EndorseRank solve that provides its restart vector; it requires 145 iterations, since a concentrated restart distribution makes it take longer to approach the stationary vector. Claim C1 is thus a statement about EndorseRank only. Any algorithm that queries the transfer layer, either alone or together with allowance information, incurs the cost of querying the transfer layer.

Two caveats apply. The baseline time measures the solver call on preconstructed edge lists; end-to-end refresh time includes parsing, decoding, and building the edge list, all of which are either common or partially common to all algorithms and thus excluded from C1. And faster is not necessarily better: the running-time result takes on its significance in light of the alignment results of Sections 4.3–4.4, which is why C4 combines the two.

4.2.1 Node-count sensitivity on the expanded pool

Table 4.2 presents runtimes at three progressive node counts taken from the expanded pool using deterministic SHA256 subsampling, along with the per-stage number of nodes and edges for each algorithm’s subgraph. The in-cohort stages ($n \leq 5,521$) appear in Appendix 8.1 (Table 8.1).

Table 4.2: Runtime versus evaluation-set size on the expanded wallet pool ($N = 99,080$; SHA256 subsampling seed `benchmark-tier2-v1`). Each stage keeps the edges that touch at least one sampled wallet, so $|V|$ and $|E|$ are reported per stage; the pool adds addresses but no additional extracted edges beyond the matched-cohort subgraph, hence the full-pool row reproduces the matched-cohort graph. Speedup = AWP/ER runtime ratio within the same row. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
10,000	0.070	0.854	12.2×	1,296	2,055	8,489	26,732
50,000	0.334	2.706	8.1×	4,440	9,190	19,990	92,887
99,080	0.401	4.105	10.2×	6,442	14,727	30,791	137,087

Two aspects of Table 4.2 should be interpreted in conjunction with Section 3.12.1. First, the full-

pool row has identical values to the matched-cohort row of Table 4.1. This is intentional and not an error: the extra addresses do not have any approved or transferred logs parsed, so the subgraph induced at $N = 99,080$ is precisely the cohort subgraph, and our benchmarking procedure times each unique graph once per session and then repeats that timing when the same graph is tabulated again. The same measurement also appears in the $d = 0.85$ row of Table 4.8. Second, the column of $|E|$ values is not monotonic in n , because each stage consists of the subgraph induced by whichever wallets are chosen according to the hash function; the stage at $n = 10,000$ happens to pick fewer cohort wallets and therefore fewer edges. What the table allows us to assert is a bounded result: within a factor of five to seven in edge count, the ratio of runtimes between AWP and EndorseRank is around $8\text{--}12\times$ and the runtime of EndorseRank is well under one-half second. It does not demonstrate the scaling of either algorithm as the number of edges increases with the number of addresses because the second-stage extraction process which was responsible for the increase in edges did not finish. Figure 4.1 illustrates both the in-cohort and pool stages on the same log axis.

4.3 Same-window alignment: EndorseRank vs. AWP (Claims C2–C3)

Claims C2 and C3 inquire about whether each method recovers its respective construct on same-window local statistics, and whether the winning methods recover their own respective constructs. Table 4.3 presents family-mean Kendall τ with 95% confidence intervals using a paired bootstrap for the transfer, allowance, and Sybil-stability families, along with the inter-method agreement between the two scoring vectors. Figure 4.2 displays these family means in a heatmap.

Table 4.3: Family-mean Kendall τ with bootstrap confidence intervals on the matched cohort ($n = 5,521$). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Family	ER mean τ	ER 95% CI	AWP mean τ	AWP 95% CI
Transfer	0.333	[0.314, 0.353]	0.534	[0.522, 0.547]
Allowance	0.355	[0.322, 0.384]	-0.030	[-0.045, -0.017]
Sybil	0.233	[0.214, 0.253]	0.286	[0.274, 0.300]
EndorseRank vs. AWP score agreement: $\tau = 0.316$, CI [0.298, 0.336]				

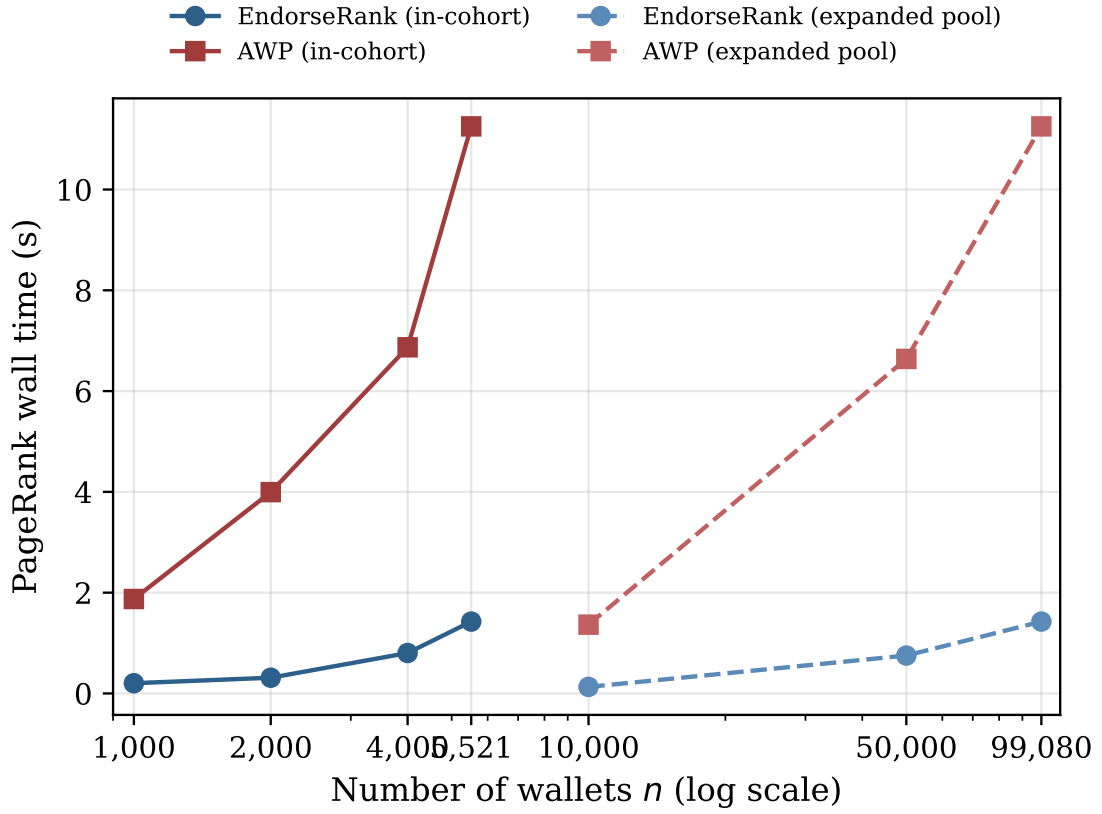


Figure 4.1: PageRank wall-clock runtime vs. number of sampled wallets (log x -axis). Solid lines: in-cohort stages ($n \leq 5,521$, Table 8.1). Dashed lines: expanded-pool stages ($N = 99,080$, Table 4.2). The pool stages add addresses but do not add any edges from extraction, so the rightmost pool point is at the full matched cohort.

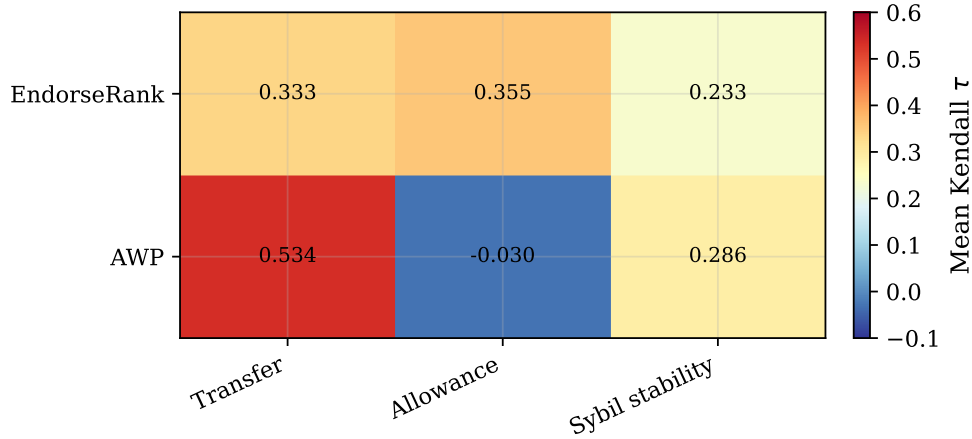


Figure 4.2: Family-mean Kendall τ on the matched cohort ($n = 5,521$) for EndorseRank and AWP; these are the values of Table 4.3.

The pattern is crossed. AWP most closely correlates with the transfer family and is slightly negative on the allowance family; EndorseRank most closely correlates with the allowance family and moderately correlates with the transfer family. Whether each of these contrasts is significant beyond sampling variation is tested in Table 4.4, which reports the five pre-specified single-layer paired differences of Section 3.10; the hybrid contrasts are in Table 4.12 (Section 4.7).

Table 4.4: Paired bootstrap tests of differences between Kendall τ coefficients (matched cohort, $n = 5,521$). Family entries use the family-mean τ ; the same wallet resamples are used for a and b , so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.355	0.333	+0.022	[-0.026, 0.067]	0.810
AWP transfer minus EndorseRank transfer	0.534	0.333	+0.201	[0.182, 0.218]	1.000
EndorseRank allowance minus AWP allowance	0.355	-0.030	+0.385	[0.350, 0.421]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.286	0.233	+0.052	[0.038, 0.065]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.356	0.344	+0.012	[-0.041, 0.060]	0.675

Three of the five contrasts have intervals that exclude zero: AWP’s transfer alignment exceeds EndorseRank’s, EndorseRank’s allowance alignment exceeds AWP’s, and AWP exceeds EndorseRank on the Sybil-stability family. Two do not. EndorseRank’s allowance alignment is not statis-

tically higher than its own transfer alignment (contrast i), and EndorseRank’s in-approve-degree alignment is not statistically higher than its in-degree alignment (contrast v). These two null results bound the interpretation of the allowance family. EndorseRank is a global smoothing of spender in-approve degree, so its agreement with allowance proxies is an intended-construct check, not an external test. Transfer centrality (AWP) carries no information about who receives spending authorization, whereas endorsement centrality (EndorseRank) does, and the reverse holds for transfer volume. The within-method claim that EndorseRank prefers allowance over transfer signal is not supported.

Claim C2 is supported in the sense that each method achieves non-trivial alignment on its own construct with an interval well away from zero. Claim C3 is supported by the two between-method contrasts (ii and iii) whose intervals exclude zero in opposite directions: neither method dominates both checks.

4.4 Proxy family detail

The family means summarize per-proxy coefficients. This section reports Spearman ρ , Kendall τ and the 95% interval of τ for each proxy. Proxies are oriented so that higher values imply higher expected reputation before correlation, and the transfer proxies are those used to validate AWP in its source paper^[3].

4.4.1 Transfer family

AWP’s agreement with in-degree is the largest coefficient in the chapter, and its agreement with in-value is smaller but still far above EndorseRank’s. Both are primary-construct checks for AWP, since it ranks on the time-decayed transfer graph; the informative observation is that AWP does not collapse to in-degree: logistic decay and value weighting move a measurable share of pairwise orderings. EndorseRank’s transfer coefficients are moderate and their intervals overlap neither zero nor AWP’s intervals. Wallets that receive many transfers are somewhat more likely to be endorsed

Table 4.5: Domain alignment with transfer proxies (inbound in-degree and in-value, the validation proxies used for the AWP baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-degree	0.422	0.344	[0.324, 0.366]
In-value (sum inbound)	0.398	0.321	[0.303, 0.342]
<i>AWP</i>			
In-degree	0.791	0.617	[0.604, 0.631]
In-value (sum inbound)	0.610	0.450	[0.436, 0.467]

spenders, but the two behaviors are far from interchangeable. If a protocol’s notion of reputation is inbound economic flow, AWP is the appropriate method on this evidence.

4.4.2 Allowance family

Table 4.6: Domain alignment with allowance proxies (spender-side in-approve degree and value). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-approve degree	0.360	0.356	[0.323, 0.385]
In-approve value (sum)	0.360	0.354	[0.321, 0.383]
<i>AWP</i>			
In-approve degree	-0.037	-0.030	[-0.045, -0.017]
In-approve value (sum)	-0.037	-0.030	[-0.045, -0.017]

EndorseRank’s coefficients on in-approve degree and in-approve value are nearly equal, which indicates that on this cohort being approved by many owners and being approved for large aggregate value convey similar ranking information under latest-allowance semantics. AWP’s coefficients are slightly negative with intervals that exclude zero: transfer centrality does not recover authorization. That dissociation is the main same-window finding. If transfer-based reputation were a sufficient proxy for endorsement, AWP would show at least moderate allowance alignment; it does not, so protocols that care about which addresses are trusted to move tokens on behalf of others cannot

substitute AWP for an allowance-aware method. The caveat of Section 4.3 applies: EndorseRank’s own allowance coefficients are a construct check, and the out-of-window test of the same construct is the holdout of Section 4.8.

4.4.3 Sybil-adjusted stability family

Table 4.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
Inbound counterparty ratio	0.093	0.073	[0.046, 0.097]
Transfer tenure (days)	0.366	0.297	[0.276, 0.318]
Active months	0.381	0.330	[0.310, 0.351]
<i>AWP</i>			
Inbound counterparty ratio	-0.063	-0.061	[-0.080, -0.040]
Transfer tenure (days)	0.602	0.436	[0.421, 0.452]
Active months	0.622	0.482	[0.466, 0.499]

AWP exceeds EndorseRank on tenure and active months, as expected from time-decayed transfer edges that reward sustained activity; the paired difference on the family mean (Table 4.4, contrast iv) excludes zero. The inbound-counterparty ratio is weak for both methods and negative for AWP. This family is a robustness note on how much longitudinal transfer information each method carries, not a headline claim.

4.5 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 ask whether EndorseRank and AWP order the same wallets differently. The inter-method Kendall τ between the two score vectors, with its interval, is given in the last row of Table 4.3. The rankings are related but far from interchangeable: agreement exceeds chance because economically active wallets participate in both approval and transfer markets, yet a large fraction of pairwise orderings disagree.

Figure 4.3 shows the marginal score distributions. Both are right-skewed, as PageRank on sparse directed graphs typically is, but they differ in shape and in the number of wallets that sit at the teleportation floor. Skewness alone does not establish divergence, so Figure 4.4 plots the ranks directly: the cloud is positively sloped but wide, with substantial off-diagonal mass.

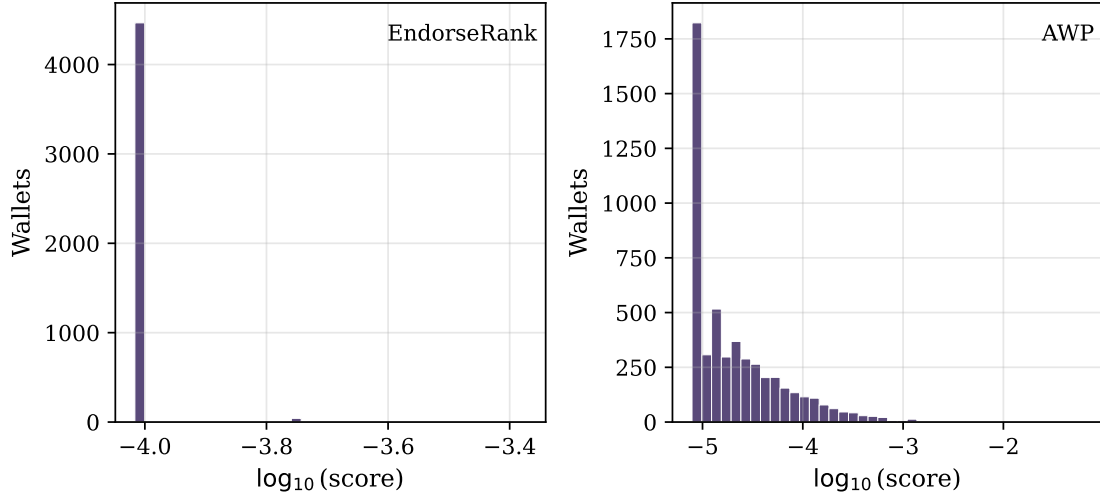


Figure 4.3: Marginal distributions of EndorseRank and AWP scores on the matched cohort ($n = 5,521$), on a $\log_{10}$ scale for wallets with positive score (EndorseRank has 966 zero-score wallets and AWP 381, omitted from the histograms).

Rank divergence is the structural counterpart of construct-specific alignment. If the two methods produced nearly identical orderings, the family-level winners of Table 4.3 could be noise around a shared ranking; the observed divergence, together with the opposite-signed between-method contrasts of Table 4.4, supports reading the two as related but distinct instruments. H1 is supported. Operationally, a protocol that currently uses transfer-based scores cannot assume that switching to, or blending in, allowance-based scores will preserve its wallet ordering; the largest disagreements are expected among transfer hubs that are rarely endorsed as spenders and among endorsed spenders with small transfer footprints.

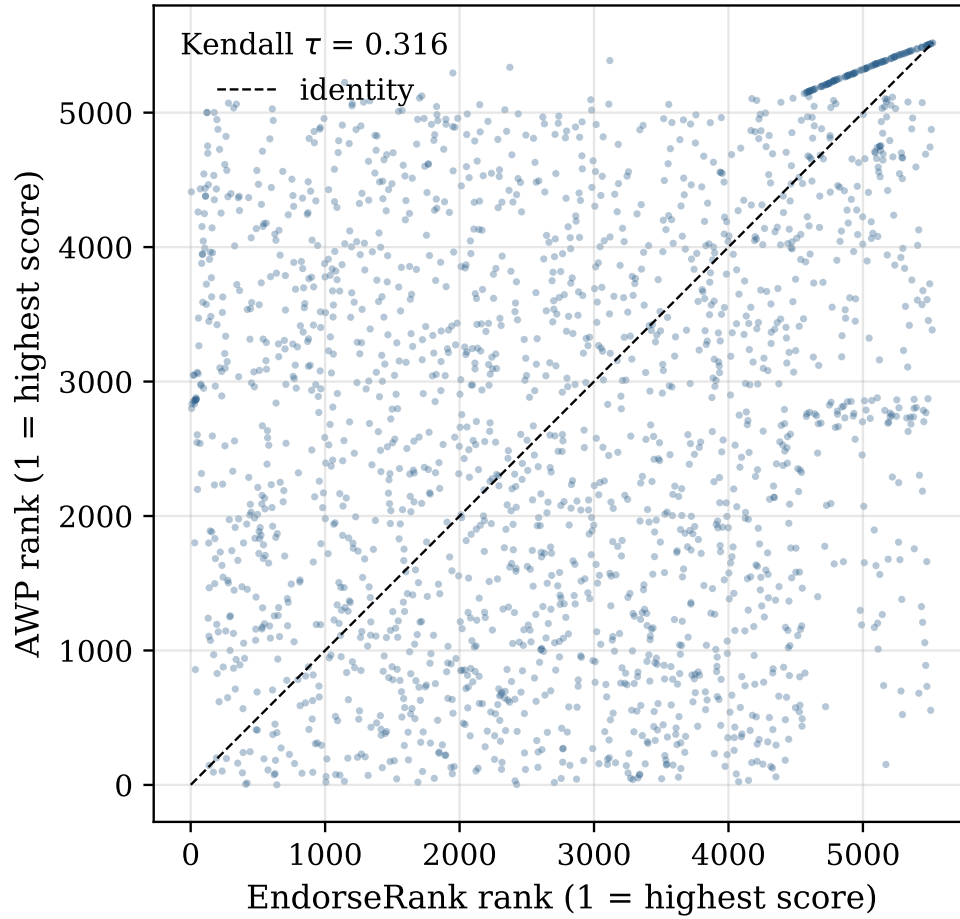


Figure 4.4: Rank–rank scatter plot of EndorseRank vs. AWP for the matched cohort (random sample of 2,000 wallets). Points on the dashed identity line represent identical orderings.

4.6 Robustness and sensitivity checks

The results above would be of limited use if they depended on a single damping value, a single token mix or a single sample definition. Three checks follow. The first two are robustness checks in the strict sense: the sample is held fixed and a parameter or the token universe is perturbed. Third, sensitivity analysis: we vary the sample definition itself, and we expect the coefficients to move.

4.6.1 Damping

Table 4.8: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched alignment cohort ($n = 5,521$). The $d = 0.85$ runtimes are the same measurements as Table 4.1 (one timing session; each graph–damping pair is timed once).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.355	-0.031	0.333	0.531	0.594	4.420
0.85	0.355	-0.030	0.333	0.534	0.401	4.105
0.95	0.355	-0.030	0.333	0.536	0.427	5.420

Family means are flat across $d \in \{0.75, 0.85, 0.95\}$ to the third decimal place for EndorseRank, and change by at most a few thousandths for AWP; the crossing of the patterns is not an artifact of the conventional $d = 0.85$. Since allowance edges represent a snapshot rather than a longer path structure, altering the amount of mass that teleports cannot sufficiently reorder wallets to impact family-level concordance. Runtimes do depend on d . AWP is substantially slower at $d = 0.95$, reflecting the slower mixing of a high-damping operator; the variation of EndorseRank over the range is of the same magnitude as its run-to-run variability in Table 4.1 and is uninterpreted. The relative runtime thus varies with damping, even though the alignment results do not. Figure 4.5 shows family means.

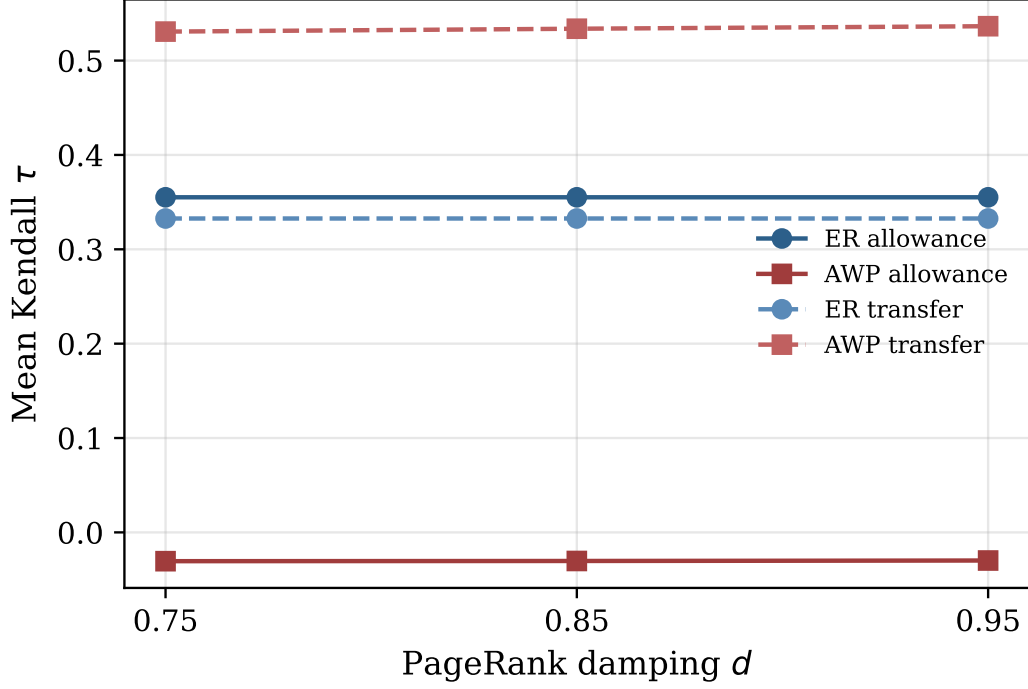


Figure 4.5: Sensitivity of family-mean Kendall τ for the allowance and transfer families over $d \in \{0.75, 0.85, 0.95\}$ (Table 4.8 values).

Table 4.9: Robustness: mean Kendall τ on the top-20 ERC-20 token subgraph (matched cohort), next to the full-token value of Table 4.3 for the same family.

Method	Transfer		Allowance		Sybil	
	top-20	full	top-20	full	top-20	full
EndorseRank	0.331	0.333	0.355	0.355	0.234	0.233
AWP	0.532	0.534	-0.031	-0.030	0.283	0.286

4.6.2 Top-token subgraph

Confining both graphs to the twenty highest-volume ERC-20 tokens changes every family mean by at most a few thousandths. Thus the alignment pattern is not driven by long-tail tokens with sparse or spam-like edges; the economically salient core of ERC-20 activity suffices to reproduce the EndorseRank–AWP contrast.

4.6.3 Sample-definition sensitivity

Table 4.10: Sensitivity of alignment to the evaluation set: mean Kendall τ and runtime at staged subsample sizes drawn from the expanded wallet pool ($N = 99,080$; seed `benchmark-tier2-v1`). Rows are evaluated on different wallet sets from the matched cohort, so τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. Runtimes are the same measurements as Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)	Speedup
10,000	0.343	0.242	0.821	0.904	0.070	0.854	12.20
50,000	0.330	0.232	0.828	0.915	0.334	2.706	8.10
99,080	0.327	0.230	0.828	0.916	0.401	4.105	10.24

Table 4.10 recomputes the transfer and allowance family means on the staged pool subsamples used for the runtime scaling. The coefficients are much larger than on the matched cohort, and AWP’s allowance coefficient turns positive. Neither change is evidence about the methods. Pool subsamples admit wallets that have no edges in one or both graphs; for such wallets the score and the proxy are both zero after alignment, and tied zeros at the bottom of both rankings raise Kendall τ mechanically (Section 3.12.4). The table is therefore reported as a sensitivity analysis of the sample definition, and it answers a narrower question: does the relative ordering of the two methods on each family survive a change of population? It does, with AWP ahead on transfer and EndorseRank on allowance at every stage, but the absolute values in the table must not be compared with those of Table 4.3, whose cohort excludes exactly the wallets that inflate them.

The damping and top-token checks support reading Claims C1–C3 as stable under parameter and token-mix perturbation. The sample-definition check shows that the comparative structure survives

a change of population while the decimal values do not.

4.7 Coupled operator: same-window results (RQ5)

Section 3.6.3 defined the coupled operator C-PR, whose transition row at each wallet mixes the allowance and transfer layers with weight λ , and the endorsement-seeded walk S-PR, which keeps the transfer layer and restarts from the EndorseRank distribution. Table 4.11 places both next to the two single-layer methods on the three same-window families of Table 4.3; Table 4.12 tests the eight hybrid contrasts of Section 3.10 against the single-layer method that leads each family. The out-of-window test that decides RQ5 is in Section 4.8; this section reports where the hybrids sit on the checks that the single-layer methods were built to pass.

Table 4.11: Same-window family-mean Kendall τ with bootstrap intervals for the two single-layer methods, the coupled operator at three values of λ and the endorsement-seeded walk (matched cohort, $n = 5,521$). $\lambda = 1$ would reproduce EndorseRank and $\lambda = 0$ AWP on a shared node set. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.333 [0.314, 0.353]	0.355 [0.322, 0.384]	0.233 [0.214, 0.253]
AWP	0.534 [0.522, 0.547]	-0.030 [-0.045, -0.017]	0.286 [0.274, 0.300]
C-PR ($\lambda = 0.25$)	0.531 [0.520, 0.545]	-0.009 [-0.020, 0.004]	0.288 [0.276, 0.302]
C-PR ($\lambda = 0.5$)	0.527 [0.515, 0.540]	-0.001 [-0.013, 0.013]	0.290 [0.279, 0.304]
C-PR ($\lambda = 0.75$)	0.519 [0.507, 0.533]	0.006 [-0.007, 0.020]	0.292 [0.281, 0.306]
S-PR	0.494 [0.481, 0.508]	0.116 [0.097, 0.134]	0.220 [0.207, 0.235]

C-PR is like AWP with a small correction for the allowance layer. On the transfer family its coefficient falls from 0.531 at $\lambda = 0.25$ to 0.519 at $\lambda = 0.75$, all within a few thousandths of AWP’s 0.534; at the primary setting the paired difference to AWP is -0.007 with an interval that excludes zero but is narrower than the rounding of the table. On the allowance family C-PR is indistinguishable from zero at every λ , far below EndorseRank’s 0.355 ($\Delta\tau = -0.356$, interval $[-0.388, -0.323]$). This happens because of the sparsity of the allowance layer. The transfer layer has about nine times as many edges as the allowance layer, and 30,791 nodes compared to 6,442 (Table 4.1). Thus most

Table 4.12: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the single-layer method that leads each family (matched cohort, $n = 5,521$). Family-mean τ ; the same wallet resamples are used for a and b . The secondary criterion of Section 3.7 is met on a family when the interval includes 0 or lies above it (transfer, allowance) or lies above 0 (Sybil stability). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
C-PR transfer minus AWP transfer	0.527	0.534	-0.007	[-0.009, -0.005]	0.000
C-PR allowance minus EndorseRank allowance	-0.001	0.355	-0.356	[-0.388, -0.323]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.290	0.286	+0.004	[0.003, 0.006]	1.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.290	0.233	+0.057	[0.043, 0.069]	1.000
S-PR transfer minus AWP transfer	0.494	0.534	-0.040	[-0.049, -0.030]	0.000
S-PR allowance minus EndorseRank allowance	0.116	0.355	-0.239	[-0.263, -0.215]	0.000
S-PR sybil-stability minus AWP sybil-stability	0.220	0.286	-0.065	[-0.072, -0.058]	0.000
S-PR sybil-stability minus EndorseRank sybil-stability	0.220	0.233	-0.013	[-0.025, -0.003]	0.007

nodes of the union graph have no allowance out-edge. For such nodes the mixing rule of Equation (3.6) uses the transfer row alone whatever the value of λ , and λ acts only at the minority of nodes that have out-edges in both layers. After the walk departs that minority it moves on transfer edges until it teleports. Raising λ to 0.75 moves the allowance coefficient by less than one hundredth. The convex mixture does not reach the allowance construct until λ is close to one, at which point it is EndorseRank.

The Sybil-adjusted stability family is the one place where C-PR exceeds both single-layer methods, as the secondary criterion of Section 3.7 asked: 0.290 against 0.286 for AWP and 0.233 for EndorseRank, with both paired intervals above zero ([0.003, 0.006] and [0.043, 0.069]). The increment over AWP is four thousandths. It is a difference in the statistical sense and not in any operational one.

S-PR is worse than the method it borrows from on every family. Seeding the transfer walk from the EndorseRank distribution lowers the transfer coefficient to 0.494 ($\Delta\tau = -0.040$ against AWP), lifts the allowance coefficient only to 0.116 ($\Delta\tau = -0.239$ against EndorseRank) and reduces Sybil-adjusted stability below both single-layer methods. A restart vector changes where the walk begins; on a graph of this density it does not change where the walk spends its time. That is the

same-window form of the finding that Section 4.8 reaches out of window.

Against the secondary criterion, C-PR at $\lambda = 0.5$ passes two of its three parts and fails the third. Its transfer coefficient (0.527) lies inside AWP’s interval $([0.522, 0.547])$; its Sybil-adjusted stability exceeds both single-layer methods with intervals above zero; its allowance coefficient (-0.001) lies far outside EndorseRank’s interval $([0.322, 0.384])$. The criterion is therefore not met. What the same-window tables show is that the mixture is dominated by the denser layer. Whether it carries any predictive content that the single layers lack is the question the holdout answers.

4.8 Temporal holdout (RQ4, RQ5)

The same-window checks cannot separate a method from its own construct. The holdout asks instead whether centrality measured up to a cut-off predicts behavior after it. Scores are frozen at 28 February 2026 on the spender cohort and two labels are counted on each wallet between 1 March and 31 May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct), following Section 3.7. Table 4.13 reports every score computed at t_1 , including the two raw-degree baselines, against both labels; Table 4.14 reports the eight pre-registered paired contrasts.

Three findings follow, in the order of the contrasts.

First, the comparison between the single-layer methods on the endorsement label stands. EndorseRank’s coefficient on new approval pairs ($\tau = 0.479$) is larger than its same-window allowance coefficient, and its paired difference to AWP ($\Delta\tau = +0.435$, interval $[0.365, 0.511]$) is the largest in the chapter; AWP’s own interval on this label includes zero. The labels occur after all scored events, hence this is out-of-window evidence that endorsement centrality contains information about future endorsement not contained in transfer centrality. On the flow label the ordering is the same but smaller: EndorseRank (0.372) exceeds AWP (0.270) with an interval that excludes zero ($[-0.148, -0.051]$ for AWP minus EndorseRank). On this cohort an approval typically precedes a transfer by the same counterparty, so the wallets receiving new owners receive new senders.

Table 4.13: Temporal holdout on the spender cohort ($n = 1,335$): Kendall τ between scores frozen at 28 February 2026 and two labels counted on the wallet between 1 March 2026 and 31 May 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the wallet for the first time. CI = 95% percentile bootstrap over wallets (400 paired resamples). 222 wallets received at least one new pair and 175 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.711 [0.665, 0.759]	0.404 [0.348, 0.467]
Transfer in-degree at t_1 (baseline)	0.117 [0.049, 0.191]	0.472 [0.423, 0.520]
EndorseRank	0.479 [0.427, 0.529]	0.372 [0.325, 0.424]
AWP	0.044 [-0.011, 0.098]	0.270 [0.235, 0.310]
C-PR ($\lambda = 0.25$)	0.271 [0.231, 0.314]	0.266 [0.231, 0.306]
C-PR ($\lambda = 0.5$)	0.295 [0.252, 0.336]	0.277 [0.241, 0.318]
C-PR ($\lambda = 0.75$)	0.314 [0.271, 0.357]	0.287 [0.253, 0.330]
S-PR	0.114 [0.052, 0.164]	0.264 [0.229, 0.304]

Table 4.14: Paired bootstrap differences on the spender holdout ($n = 1,335$, 400 resamples shared by a and b). The first two rows give the increment of each PageRank over its own raw degree at t_1 ; the next two compare the single-layer methods on the other construct’s label; the last four are the pre-registered hybrid contrasts. Primary criterion (C-PR no worse than EndorseRank on new approval pairs and better than AWP on new transfer senders): not met.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank minus t1 in-approve degree	new approval pairs	0.479	0.711	-0.233	[-0.291, -0.183]	0.000
AWP minus t1 in-degree	new transfer senders	0.270	0.472	-0.201	[-0.230, -0.171]	0.000
EndorseRank minus AWP	new approval pairs	0.479	0.044	+0.435	[0.365, 0.511]	1.000
AWP minus EndorseRank	new transfer senders	0.270	0.372	-0.102	[-0.148, -0.051]	0.000
C-PR minus EndorseRank	new approval pairs	0.295	0.479	-0.184	[-0.213, -0.149]	0.000
C-PR minus AWP	new transfer senders	0.277	0.270	+0.007	[-0.012, 0.028]	0.735
S-PR minus EndorseRank	new approval pairs	0.114	0.479	-0.365	[-0.431, -0.300]	0.000
S-PR minus AWP	new transfer senders	0.264	0.270	-0.006	[-0.017, 0.006]	0.163

Second, neither PageRank improves on its own raw degree at t_1 , and the second part of RQ4 is answered in the negative. In-approve degree at t_1 predicts new approval pairs with $\tau = 0.711$ versus EndorseRank’s $\tau = 0.479$ ($\Delta\tau = -0.233$, interval $[-0.291, -0.183]$); transfer in-degree at t_1 predicts new transfer senders with $\tau = 0.472$ versus AWP’s $\tau = 0.270$ ($\Delta\tau = -0.201$, interval $[-0.230, -0.171]$). Both intervals are well below zero. The number of new counterparties within a three-month window increases with the number of existing counterparties, and the local count captures that directly; PageRank replaces the count with a weighted, propagated quantity in which owner-side centrality and allowance value also enter, and on this label those extra terms serve as noise. The out-of-window predictive content of EndorseRank is contained in, and diluted from, the spender’s t_1 in-approve degree. The same holds for AWP and the transfer in-degree. Whatever either PageRank contributes beyond its degree must thus be found in same-window structure (Sections 4.3 and 4.5), not in predicting the construct’s own future growth.

Third, the coupled operator fails the pre-registered primary rule on both components. C-PR at $\lambda = 0.5$ is worse than EndorseRank on new approval pairs ($\Delta\tau = -0.184$, interval $[-0.213, -0.149]$) and no different from AWP on new transfer senders ($\Delta\tau = +0.007$, interval $[-0.012, 0.028]$). The three values of λ order monotonically on both labels: τ on new approval pairs rises from 0.271 ($\lambda = 0.25$) through 0.295 to 0.314 ($\lambda = 0.75$), and on new transfer senders from 0.266 through 0.277 to 0.287. More allowance weight helps on both labels and the transfer layer contributes no out-of-window signal for this cohort; the coupled score interpolates between the two single-layer scores rather than dominating either one. The seeded ablation explains why. S-PR equals AWP on the flow label ($\Delta\tau = -0.006$, interval $[-0.017, 0.006]$) and is far below EndorseRank on the endorsement label ($\Delta\tau = -0.365$): once the transfer structure dictates propagation, an endorsement-derived restart distribution will change very little, so the allowance edge is useful only when allowing propagation over it, not when used to seed from it.

There are two caveats that hold for the entire table. Each label is the same construction in the future (new approvals, new senders), so none of this is about credit or loss. Both labels are sparse: 222 of 1,335 wallets got a new approval pair and 175 a new sender, so each coefficient comes entirely

from ordering within that small subset.

The other candidate labels computed on the holdout are not shown as support. Revocation counts and the drain heuristic correlate positively with EndorseRank, which has the wrong sign for an endorsement interpretation (a heavily endorsed spender is also a frequently revoked one), and Aave liquidations in this spender set are empty. The research proposal’s trading-success comparison was dropped for the reason stated in Section 2.8: the outcome-native baseline it used was built from the same GMX events as the proxies it would validate against.

4.9 Primary claims (C1–C5)

This chapter’s results validate these main claims, cross-referenced to the hypothesis and research question columns in Table 1.2.

- C1. On identical wallet sets, EndorseRank solves PageRank in a fraction of AWP’s wall-clock time, using proportionally less memory and fewer iterations, since its latest-allowance graph has roughly one-ninth the edges of the transfer graph (Table 4.1). This ratio is constant across in-cohort and pool stages that differ by a factor of five to seven in edge counts (Tables 4.2 and 8.1); pool stages do not test for further increases in edge count beyond the cohort graph. This claim only applies to EndorseRank: the coupled operator runs on both edge sets and costs at least as much as AWP.
- C2. Each method recovers its target construct on same-window local statistics, with bootstrap confidence intervals that do not include zero (Table 4.3). These tests assess internal consistency, not external validity.
- C3. The winners are construct-specific: AWP is better than EndorseRank on the transfer family and vice versa on the allowance family, with paired-difference confidence intervals excluding zero in opposite directions (Table 4.4); the two score vectors only partially agree (Table 4.3). EndorseRank’s advantage on allowances compared to its own transfer-based ranking is not significantly different.

- C4. The efficiency and construct findings create a trade-off. EndorseRank is the more cost-effective method, is the only one consistent with authorization-based scoring, and on the spender holdout predicts new approvals that AWP does not (Table 4.13). But Table 4.14 bounds this out-of-sample claim: the spender’s raw in-approve degree at t_1 predicts the same label more accurately than EndorseRank, and the transfer in-degree predicts new senders better than AWP. AWP is still the superior method for transfer-hub identification and time-series stability.
- C5. Combining both edge types in a single random walk does not outperform either single-type method. For each λ tested, C-PR falls between EndorseRank and AWP on both out-of-sample labels, the pre-registered success criterion is not met, and the seeded version replicates AWP (Tables 4.13 and 4.14). The same-window position of the hybrids is reported in Section 4.7. C5 is a bounded negative result: on this chain and cohort the two edge types do not combine profitably under a per-node convex mixture with fixed λ .

RQ1 is answered by the inter-method agreement and Figure 4.4; RQ2 by the family tables and difference tests; RQ3 by the benchmark tables; RQ4 and RQ5 by the holdout. The claims do not retire AWP, and they do not promote EndorseRank beyond what the intervals allow: EndorseRank is cheaper, aligns with allowance checks, diverges moderately from AWP, predicts later new approvals where AWP does not, and is itself out-predicted by the raw degree it smooths.

Chapter 5 interprets these findings against the research questions and objectives of Chapter 1, draws out their theoretical, practical and societal implications, and examines the threats to validity and limitations that bound them.